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SUMMIT RIDGE · ML ENGINEERING

Model evaluation metrics that actually matter

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"Our model is 95% accurate" sounds impressive until you learn the model is predicting fraud, and only 1% of transactions are actually fraudulent. A model that just guesses "not fraud" every single time also hits 95% accuracy — and catches zero fraud. This is why "accuracy" alone is one of the most misleading numbers in ML.

- ◆ Precision — of everything the model flagged as positive, how much was actually right? High precision means few false alarms.
- ◆ Recall — of everything that was actually positive, how much did the model catch? High recall means few missed cases.
- ◆ These trade off against each other — a model can catch every fraud case by flagging everything (perfect recall, terrible precision), so the right balance depends entirely on the cost of each type of error in your specific use case. Missing a fraud case and annoying a legitimate customer are not equally costly.

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“What's a time a headline metric hid a real problem in your work?”

Follow along – Day 25 drops tomorrow.

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#AI

#ModelEvaluation